A Conditional Reduction for the Raw Weak-Compactness Index Cutoffs

Run fa_banach_001, agent agent_lane_18

2026-06-15

Status

This is a conditional packet, not a full solution of Question 9.3 of Beanland–Causey–Freeman–Wallis [1]. The conditional dependency is isolated in Lemma 1. If that lemma is proved, then the raw weak-compactness cutoff classes at the principal ordinals ω^{ω^η}, $\eta \geq 1$, are norm-closed operator ideals.

1 Source Question

In [1, Question 9.3], after defining the raw weak-compactness index $\mathbf{WC}(A, X, Y)$ by finite sequences in B_X whose images dominate the summing basis of c_0, the authors ask:

> For which ordinals ξ is the class of operators $A : X \to Y$ such that $\mathbf{WC}(A, X, Y) \leq \xi$ an ideal?

The source paper proves that the cutoff $\mathbf{WC} \leq \omega$ is the ideal of super weakly compact operators and that $\mathbf{WC} < \omega_1$ is an ideal. A previous packet in this run proves that finite cutoffs $n \geq 2$ are not closed under sums.

2 Definitions

Let (s_i) denote the summing basis of c_0, with

$$\left\| \sum_{i=1}^{n} a_i s_i \right\| = \max_{1 \leq m \leq n} \left| \sum_{i=1}^{m} a_i \right|.$$

For $K \geq 1$, $\mathbf{WC}(A, X, Y, K)$ is the tree of all finite sequences $(x_i)_{i=1}^n \subset B_X$ such that $(Ax_i)_{i=1}^n$ K-dominates $(s_i)_{i=1}^n$. Then

$$\mathbf{WC}(A, X, Y) = \sup_{K \geq 1} o(\mathbf{WC}(A, X, Y, K)).$$

Causey's James index [2] is built from convexly separated sequences. For a body $K \subset X^*$ and $\varepsilon > 0$, a finite sequence $(x_i)_{i=1}^n$ is (K, ε)-convexly separated if, for every $1 \leq m < n$,

$$d_K\big(\mathrm{co}\{x_i : i \leq m\}, \mathrm{co}\{x_i : m < i \leq n\}\big) \geq \varepsilon.$$

For an operator $A : X \to Y$, this is applied to $K = A^* B_{Y^*}$; equivalently, the image sequence (Ax_i) is convexly separated in norm. Let $j(A, \varepsilon)$ be the order of this tree and $\mathcal{J}(A) = \sup_{\varepsilon > 0} j(A, \varepsilon)$.

3 Elementary Comparison

Lemma 1. *For every operator $A : X \to Y$, $\mathbf{WC}(A, X, Y) \leq \mathcal{J}(A)$.*

Proof. Fix $K \geq 1$ and $(x_i)_{i=1}^n \in \mathbf{WC}(A, X, Y, K)$. Given $1 \leq m < n$, take convex combinations $u = \sum_{i \leq m} \lambda_i A x_i$ and $v = \sum_{i>m} \mu_i A x_i$. The coefficient vector $(\lambda_1, \dots, \lambda_m, -\mu_{m+1}, \dots, -\mu_n)$ has summing-basis norm at least 1, because its mth partial sum is 1. Hence

$$\|u - v\| \geq K^{-1}.$$

Thus every $\mathbf{WC}(A, K)$ node is a $J(A, K^{-1})$ node, so $o(\mathbf{WC}(A, K)) \leq j(A, K^{-1})$. Taking the supremum over K gives the claim. □

4 Conditional Dependencies

Conditional Lemma 1 (James-to-wide-(s) tree bridge)**.** *For every operator $A : X \to Y$, every $\varepsilon > 0$, and every ordinal α, if*

$$j(A, \varepsilon) > \omega\alpha,$$

then there is some $K \geq 1$ such that

$$o(\mathbf{WC}(A, X, Y, K)) > \alpha.$$

This is the only unproved dependency in the packet. It is plausible for the following reasons. First, Causey's $J(A, \varepsilon)$ trees are closed under successive positive convex blocks: a positive convex block of a convexly separated branch remains convexly separated. Second, BCFW use an $\omega\alpha \mapsto \alpha$ tree-blocking argument to make branches basic in related Bourgain trees. Third, Rosenthal's finite theory of λ-wide-(s) sequences [3] identifies the finite quantitative version of domination of the summing basis and proves finite triangular-array extraction results. A proof of Lemma 1 should combine these three ingredients.

The dependency is not automatic. Convex separation alone does not imply summing-basis domination at the same finite level; finite-dimensional monotone scalar configurations can be convexly separated while failing even length-two raw summing domination. The expected loss of one factor of ω is meant to remove precisely this finite-dimensional affine obstruction.

5 Idea of the Proof

The key observation is that the raw WC index is bounded above by Causey's James index, while Lemma 1 gives a partial converse after one ω loss. At ordinals $\gamma = \omega^{\omega^\eta}$ with $\eta \geq 1$, this loss disappears, since $\omega\gamma = \gamma$. Therefore the two ranks have the same cutoff at these principal ordinals. Causey's theorem then transfers ideality for the James cutoff to the raw WC cutoff, and the elementary comparison transfers the sum back.

6 Conditional Theorem

Theorem 1. *Assume Lemma 1. Let $\eta \geq 1$ and put $\gamma = \omega^{\omega^\eta}$. Then the class*

$$\mathcal{W}_\gamma = \{A : X \to Y : \mathbf{WC}(A, X, Y) \leq \gamma\}$$

is a norm-closed operator ideal.

Proof. We first show that $\mathbf{WC}(A) \le \gamma$ implies $\mathcal{J}(A) \le \gamma$. If not, then for some $\varepsilon > 0$,

$$j(A,\varepsilon) > \gamma.$$

Since $\eta \ge 1$, $\gamma = \omega^{\omega^\eta}$ is fixed by left multiplication by ω, i.e. $\omega\gamma = \gamma$. Thus $j(A,\varepsilon) > \omega\gamma$, and Lemma 1 gives $o(\mathbf{WC}(A,K)) > \gamma$ for some K, contradicting $\mathbf{WC}(A) \le \gamma$. Hence

$$\mathcal{W}_\gamma \subseteq \{A : \mathcal{J}(A) \le \gamma\}.$$

Causey proves that $\{A : \mathcal{J}(A) \le \omega^{\omega^\eta}\}$ is a closed operator ideal for every ordinal η [2]. Let $A, B : X \to Y$ lie in $\mathcal{W}_\gamma$. By the previous paragraph, $\mathcal{J}(A), \mathcal{J}(B) \le \gamma$, so Causey's theorem gives $\mathcal{J}(A+B) \le \gamma$. Lemma 1 then gives

$$\mathbf{WC}(A+B) \le \mathcal{J}(A+B) \le \gamma.$$

Thus $\mathcal{W}_\gamma$ is closed under finite sums.

Closure under composition follows directly from the definition of $\mathbf{WC}$: if $C : W \to X$ and $D : Y \to Z$ are bounded, any raw WC tree for DAC rescales to a raw WC tree for A, with only the domination constant changed. Therefore $\mathbf{WC}(DAC) \le \mathbf{WC}(A)$.

Finally, if $A_n \to A$ in norm and $A_n \in \mathcal{W}_\gamma$, then $\mathcal{J}(A_n) \le \gamma$ by the first paragraph. Causey's class $\{T : \mathcal{J}(T) \le \gamma\}$ is norm closed, so $\mathcal{J}(A) \le \gamma$. Lemma 1 gives $\mathbf{WC}(A) \le \gamma$. Hence $\mathcal{W}_\gamma$ is norm closed. □

7 Verification Notes

The cheap run indexes were searched for arXiv:1507.06285, "weak compactness index", "summing basis", "Causey", "James index", and "convexly separated". Existing run memory contains the finite-cutoff partial result and a prior full-solution push attempt, but no infinite-cutoff solution.

The bounded literature search for this packet found:

- Causey's arXiv:1508.02065 theorem that the James-index cutoffs $\mathcal{J} \le \omega^{\omega^\eta}$ are closed operator ideals.
- Rosenthal's arXiv:math/9610209 finite and infinite theory of λ-wide-(s) sequences, triangular arrays, and super weak compactness.
- No source in the local index or web search explicitly answering BCFW Question 9.3 for the raw **WC** cutoffs beyond the source paper's $\mathbf{WC} \le \omega$ result.

8 Limitations

This packet does not prove Lemma 1. Without that bridge, Theorem 1 remains conditional. The most likely obstruction is that convex-separation trees may contain high-rank affine configurations that do not admit a uniform raw summing-basis subtree after one ω loss. A counterexample should probably come from a well-founded James-tree-type reflexive space with large Causey $\mathcal{J}$ rank but unexpectedly small raw **WC** rank.

References

[1] K. Beanland, R. Causey, D. Freeman, and R. Wallis, *Classes of operators determined by ordinal indices*, arXiv:1507.06285.

[2] R. M. Causey, *An ordinal index characterizing weak compactness of operators*, arXiv:1508.02065.

[3] H. P. Rosenthal, *On wide-(s) sequences and their applications to certain classes of operators*, arXiv:math/9610209.